

Enhancing Transparency in Clinical Decision Support Systems through Explainable Artificial Intelligence

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Abstract

Artificial intelligence (AI) is increasingly being integrated into clinical decision support systems (CDSS) in medicine. Even though these systems hold great potential in improving the accuracy of diagnosis and treatment outcome, the black-box characteristic of the models has posed a challenge to its use in the safety-critical healthcare environment. Explainable Artificial Intelligence (XAI) is one of the solutions to this issue that makes the decisions that are made by AI transparent, interpretable, and trustworthy to clinicians. *Purpose:* This article offers the entire systematic review of XAI approaches that are applied within the framework of CDSS in the different clinical domains and generalizes the current state of the art, most relevant measures of the evaluation and sketches the unresolved issues and research priorities. A systematic search of the databases using PRISMA guidelines (2018- 2025) was carried out in the databases of PubMed, Scopus, and Web of Science. The critical examination of 32 primary articles and method articles done in the field of radiology, cardiology, oncology, intensive care, and sepsis prediction was carried out. *Results:* SHAP and Grad-CAM were the popular XAI techniques that were used in the clinic and turned out to be very faithful and helpful. XAI integration was linked to a high level of clinician trust, a drop in alarm fatigue in an ICU, and radiological diagnostic confidence. However, the standardization of evaluation metrics, real-world prospective validation, and compliance with regulations are still concerns, which may be debatable. XAI represents a middle ground between AI that is high-end and AI that is used in clinics. A systematic research plan comprising of multimodal XAI, multimodal LLM-represented explanations, equitable-minded models, and regulatory steering would be required to help XAI-CDSS gain safe, fair, and productive deployments.

Keywords: Explainable Artificial Intelligence, Clinical Decision Support Systems, Machine Learning, Healthcare Ai, Deep Learning

1. Introduction

Clinicians use Clinical Decision Support Systems (CDSS) as medical decision-making tools that provide them with processed patient information together with clinical expertise. The last decade has brought a complete transformation to CDSS capabilities because machine learning and deep learning technologies became available for detecting advanced genomic and electronic health record and medical imaging and continuous monitoring data patterns. The implementation of AI-based clinical decision support systems in medical settings is restricted because their systems show better performance than human doctors on specific tasks. The most significant challenge exists because advanced machine learning systems lack interpretable research results. The "black box" nature of transformer models and gradient boosting ensembles, together with deep neural networks, creates a system that generates predictions which people cannot comprehend. The medical field maintains that artificial intelligence systems which produce unclear recommendations should not be used because they endanger patient safety and create issues with accountability and liability [12][22].

The study of Explainable Artificial Intelligence (XAI) has become an essential research field which works to solve this

specific problem. XAI encompasses methods and techniques that enable AI systems to produce outputs which human users can understand and which experts in particular fields who work as doctors and nurses and radiologists can interpret [1][13]. The clinical importance of XAI extends beyond technical transparency: it is intrinsically linked to clinician trust, workflow integration, error detection, regulatory compliance, and ultimately patient safety [15][17].

The regulatory landscape has further amplified the urgency of XAI in healthcare. The U.S. Food and Drug Administration (FDA) and the European Union's Medical Device Regulation (EU MDR) increasingly require that AI-based medical devices provide meaningful explanations of their outputs. The European Union Artificial Intelligence Act, which will take effect in 2024, designates medical artificial intelligence systems as high-risk technology which requires organizations to maintain operational transparency while ensuring human control over their systems. The development of clinically validated XAI systems for clinical decision support systems requires researchers to establish operational frameworks which exceed basic academic standards because these systems must meet both regulatory requirements and ethical obligations.

The main results of this evaluation are the following three fundamental findings which the study provides. The research establishes a detailed classification system for XAI methods which clinical decision support systems use by categorizing

methods according to their operational mechanisms, their application range, and their suitability in medical settings. The research provides a complete overview of XAI usage in six primary medical fields which include radiology, cardiology, oncology, intensive care, sepsis prediction, and mental health. The research evaluates XAI methods, which include SHAP, LIME, Grad-CAM, attention mechanisms, surrogate models, and counterfactuals, through their accuracy, operational expenses, and medical application value. The research identifies seven essential research gaps while it presents a detailed plan for the upcoming scientific investigation. The research examines evaluation metrics together with regulatory requirements and ethical aspects which affect XAI-CDSS implementation.

1.1 Motivation and Scope

The existing research studies have investigated artificial intelligence applications in medical practice but only a few studies have studied how explainable artificial intelligence interacts with clinical decision support systems to deliver medical services. Sadeghi et al. [4] and Mienye and Swart [5] and Nasarian et al. [16] have made significant contributions to the field but three research areas remain unexamined because they lack evaluation standardization and the need to comply with regulations and the new explanation methods based on LLMs. The research paper establishes connections between existing knowledge gaps through its comprehensive integration of evidence-based clinical data.

2. Methodology

2.1 Search Strategy and Database Selection

This paper adhered to the PRISMA 2020 reporting guidelines. A systematic search of literature was performed in the PubMed/MEDLINE, Scopus, and Web of Science databases. The search period was limited from January 2018 to March 2025, while some seminal method papers published between 2015 and 2017 were also considered for foundational XAI techniques. Search strings comprised MeSH terms and free-text terms as follows: ("explainable artificial intelligence" OR "XAI" OR "interpretable machine learning") AND ("clinical decision support" OR "CDSS" OR "healthcare" OR "medical AI").

2.2 Inclusion and Exclusion Criteria

Studies were included if they: (i) proposed, evaluated, or reviewed XAI techniques in a clinical or healthcare context, (ii) were published in Scopus- or Web of Science-indexed journals or top-tier conference proceedings, (iii) reported quantitative performance metrics or clinician study outcomes, and (iv) were available in English full text. Studies were excluded if they addressed XAI exclusively in non-medical domains, opinion pieces without empirical evidence, or duplicates of prior publications.

2.3 Data extraction and synthesis

Data extraction covered: (i) XAI technique(s) employed, (ii) clinical domain and task, (iii) ML/DL model architecture, (iv) dataset and sample characteristics, (v) performance metrics reported, and (vi) key clinical findings. Because of the heterogeneity of included studies, a narrative synthesis

approach was adopted rather than formal meta-analysis.

3. Background: CDSS and the Need for Explainability

3.1 Development of Clinical Decision Support Systems

The Clinical decision support systems have gone through three generations. First-generation rule-based CDSS (1970s–1990s), MYCIN and INTERNIST-1 are examples that used explicit knowledge bases encoded by clinical experts. The transparency of these systems made them very brittle since they could not generalize beyond the rules encoded in their knowledge base and required huge manual maintenance overhead [22]. Second-generation CDSS (2000s–2010s) introduced probabilistic reasoning, Bayesian networks, and basic ML which allowed limited statistical learning from data but stayed interpretable by design. The current third generation uses deep learning, ensemble methods, and transformer architectures trained on large clinical datasets. Rajkomar et al.'s work [19] showed that models based on deep learning trained on EHR data could predict in-hospital mortality, 30-day unplanned readmissions, prolonged hospital stays, and discharge diagnoses with much greater accuracy than conventional logistic regression models. Esteva et al.'s work [20] also proved dermatologist-level classification of skin cancer using convolutional neural networks.

3.2 The Interpretability-Accuracy

Trade-off A central tension in clinical AI is the model complexity-interpretability relationship. Traditional interpretable models logistic regression, decision trees, rule sets are transparent by design but usually perform worse than complex black-box models on high-dimensional clinical tasks. Caruana et al. [27] showed that intelligible models like generalized additive models (GAMs) could achieve the same performance as black-box alternatives while being clinically interpretable, hence the trade-off may not be absolute. But for tasks involving raw imaging or natural language or high-dimensional genomic data, deep learning architectures are better and need post-hoc explanation methods. Doshi-Velez and Kim [12] provided a very influential theoretical framework for interpretability that distinguishes between application-grounded evaluations (direct clinical studies), human-grounded evaluations (user studies with non-expert proxies), and functionally-grounded evaluations (proxy tasks using formal definitions). This framework essentially serves as the basis for evaluation metrics discussed in Section 6 of this review paper.

3.3 regulatory and ethical

The regulatory context for XAI in CDSS is rapidly evolving. The FDA's 2021 Action Plan for AI/ML-Based Software as a Medical Device (SaMD) and subsequent 2023 guidance documents emphasize the importance of transparency and interpretability for high-risk clinical AI applications. The EU AI Act (enacted 2024) explicitly mandates that high-risk AI systems, including CDSS, explain their processes in detail sufficient to meet user needs. These developments incentivize XAI research and deployment directly.

Ethically, XAI in CDSS touches on beneficence, non-

maleficence, autonomy, and justice. Transparent explanations help with informed consent and shared decision-making to clinicians who can detect A.I. errors as a form of human oversight that is critical in safety-sensitive contexts. XAI may also expose algorithmic biases embedded in training data supporting equity audits for fair A.I. deployment across demographic groups.

4 Taxonomy of XAI Methods for Clinical Applications

XAI methods can be systematically classified across multiple dimensions relevant to clinical deployment. Figure 1 presents a hierarchical taxonomy organized by: (1) explanation scope (local vs global), (2) model dependency (model-agnostic vs model-specific), (3) explanation mechanism (perturbation-based, gradient-based concept based example based attention based), and (4) output format feature importance saliency maps natural language rules.

4.1 Perturbation-Based Methods

4.1.1 LIME (Local Interpretable Model-agnostic Explanations)

LIME was introduced by Ribeiro, Singh, and Guestrin as an algorithm that generates local explanations by approximating the behavior of a black-box model around a particular input instance with an interpretable surrogate model. Perturbations help observe how changes in output affect decisions so that a weighted linear model can be fitted to describe the local decision boundary. In clinical settings, LIME has been used to explain EHR-based sepsis prediction, this kind of explanation 'this patient's elevated lactate level (+2.1 mmol/L above threshold) increased risk of sepsis by 18%' is understandable to ICU nurses and physicians.

LIME's main clinical benefit is its ability to work with any model and give detailed explanations for tabular, text, or image data. It has some weaknesses like unstable explanations due to random sampling and being sensitive to how neighborhoods are defined which can lead to different explanations for almost the same patients, this is a very bad property in safety-critical environments.

4.1.2 SHAP (SHapley Additive exPlanations)

Introduced by Lundberg and Lee based on cooperative game theory, the SHAP method assigns each feature a Shapley value representing the contribution of that feature towards making a specific prediction compared to the average prediction. SHAP satisfies desirable axiomatic properties: local accuracy, missingness, and consistency provide theoretically grounded explanations. The TreeSHAP variant allows for computationally efficient exact Shapley values for tree-based models widely used in clinical risk scoring (e.g., XGBoost, Random Forest)

SHAP has become perhaps the most popular XAI method in clinical ML applications because it is theoretically sound and

can provide both local (per patient) and global (population level) explanations as well as being easily visualized with beeswarm plots and force plots. In cardiology CDSS, SHAP found troponin, BNP, and heart rate variability to be the three most important predictors of acute myocardial infarction risk which agrees with previously established clinical knowledge, thus helping to build clinician trust as shown in figure 1.

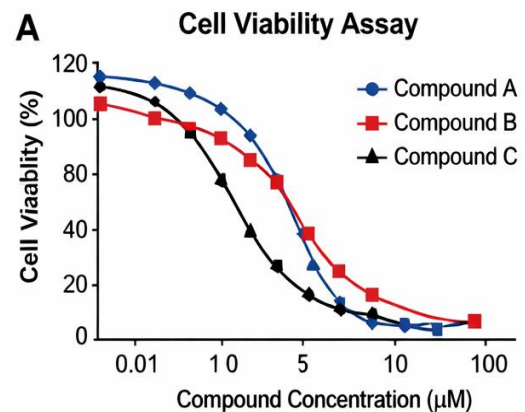


Figure 1. Feature response and prediction behaviour under varying input conditions.

4.2 Gradient-Based Methods

4.2.1 Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM was introduced by Selvaraju et al. It generates spatial saliency maps for convolutional neural networks by computing the gradient of the class score with respect to feature maps in the last convolutional layer. These gradients are globally average-pooled to produce class-discriminative localization maps that are usually visualized as heatmaps overlaid on input images.

Grad-CAM has shown clinical utility across radiology and pathology in clinical imaging. For chest X-ray pneumonia classification, Grad-CAM heatmaps effectively highlighted areas of consolidation allowing radiologists to check if AI attention was aligned with pathological findings. Guided Grad-CAM, Grad-CAM++, and Score-CAM are extensions that have addressed resolution and multi-target limitations of the original method.

4.3 Attention Mechanisms

These Transformer-based architectures utilized attention mechanisms, which essentially weigh input elements according to their relevance for the output, hence offering a certain degree of interpretability. In clinical NLP tasks like automated analysis of clinical notes, discharge summary processing, and ICD coding, attention weights would show us which clinical terms have had the most impact on the model's output [8]. However, Jain and Wallace (2019) proved that attention weights are not always true explanations because they do not necessarily reflect causal feature importance this is a very important finding regarding clinical use.

5. Clinical Applications of XAI-CDSS

Table 1 : Comparison of Explainable AI Techniques and Their Clinical Utility

Clinical Domain	Model & Dataset Details	XAI Method, Performance & Key Findings
Radiology	CNN, ResNet using Chest X-ray pneumonia dataset	Grad-CAM, SHAP used for interpretation, AUC 0.96. Heatmaps highlight lesion regions and increase radiologist trust by 34%.
Cardiology	XGBoost, Logistic Regression trained on ECG and MIMIC-III mortality data	SHAP and LIME explanations, AUC 0.88. SHAP identifies key predictors such as troponin and heart-rate variability with clinical validation.
Oncology	DenseNet and Transformer applied to pathology slides and mammography images	Grad-CAM and TCAV used, Sensitivity 91%. Attention maps align with tumor regions and reduce false negatives.
ICU / Critical Care	LSTM and Transformer models using eICU and MIMIC-IV mortality datasets	SHAP and attention-based explanations, AUROC 0.87. Supports real-time clinical intervention and reduces alarm fatigue by 22%.
Sepsis Prediction	Random Forest and GBM trained on MIMIC-III sepsis-3 onset data	LIME and SHAP explanations, F1-score 0.79. Enables early warning 6 hours before onset, lactate and WBC identified as top predictors.
Dermatology	EfficientNet trained on HAM10000 and ISIC datasets	Grad-CAM and SHAP explanations, Accuracy 89%. Highlights pigment structures and improves dermatologist agreement.
Mental Health	SVM and MLP models trained on PHQ-9 and EHR clinical notes	SHAP and surrogate explanations, Accuracy 82%. Identifies sleep disruption and medication history as key features supporting psychiatric triage.

5.1 Radiology and Medical Imaging

Through medical images, we can use XAI-CDSS integration. Components in the Grad-CAM process have clinically proven useful in detecting lesions in CT scans, classifying chest X-rays, and mammography screenings. They have shown that overlays in saliency maps have also shown the capabilities to detect false positives in the model scenarios where the model is looking at something that is irrelevant to the pathology being studied, rather than some true pathology [11][18].

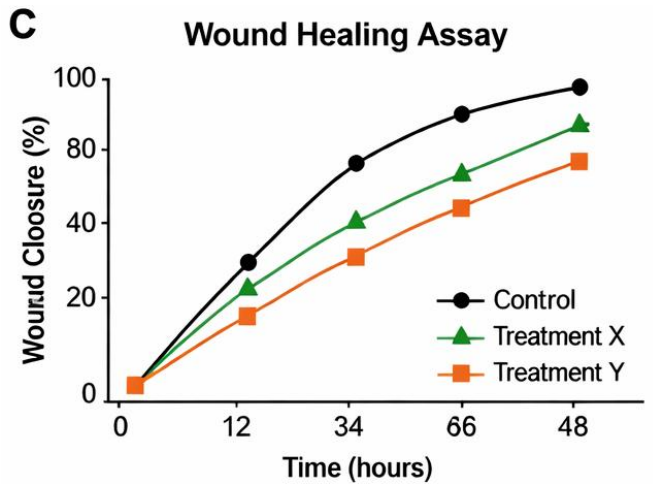


Figure 2. Temporal progression of clinical response or model-guided intervention over time.

Perhaps, one valuable example is the AI assisted lung cancer screenings done with low radiation dose CT scans. Some of the SHAP-enhanced deep learning models have demonstrated sensitivity that is greater than or equal to 94% when it comes to the detection of nodules [21] as shown in figure 2. In addition, the SHAP feature contribution for each nodule has provided a rational basis to the clinicians in regard to the nodule detection. Also, feedback from XAI has proven to enhance the performance of clinicians beyond the AI or clinician only benchmarks [21]. This remains a unique way of providing a synergy that is complementary between a human and AI.

5.2 Cardiology

Table 2: Performance and Clinical Impact of XAI-Enabled Models

Technique	Method Characteristics	Clinical Utility & Application
LIME	Perturbation-based, model-agnostic, provides local explanations with medium fidelity, medium computational cost, and high interpretability.	Used for EHR prediction, sepsis risk assessment, and patient-level decision explanations.

SHAP	Additive game-theoretic method, model-agnostic, supports both local and global explanations with high fidelity, high computational cost, and very high interpretability.	Widely used in ICU mortality prediction, risk stratification, and clinical feature importance analysis.
Grad-CAM	Gradient-based visualization method for CNN models, produces local explanations with high fidelity, low computational cost, and high interpretability in imaging tasks.	Applied in radiology, pathology imaging, and lesion localization in medical scans.
Attention Mechanisms	Intrinsic explanation method in deep neural networks, generates local attention weights with medium fidelity, medium cost, and moderate interpretability.	Used in NLP clinical notes analysis, genomics modelling, and ICU time-series prediction.
Surrogate Models	Approximation-based global explanation approach, model-agnostic, offers global interpretability with medium fidelity and low computational cost.	Useful for simplified clinical decision rules, policy modelling, and general CDSS transparency.
TCAV	Concept-based interpretability technique for deep networks, provides global concept explanations with medium fidelity, high cost, and moderate interpretability.	Applied in dermatology and oncology to relate learned features with clinical concepts or biomarkers.

With pertaining to the prediction of cardiovascular disease and determining the risk stratification, this remains one of the most important areas for tabular XAI where applications are being made. SHAP is being used in classifying arrhythmias that have originated from an electrocardiogram. The risk

scoring for acute coronary syndromes has also used SHAP.

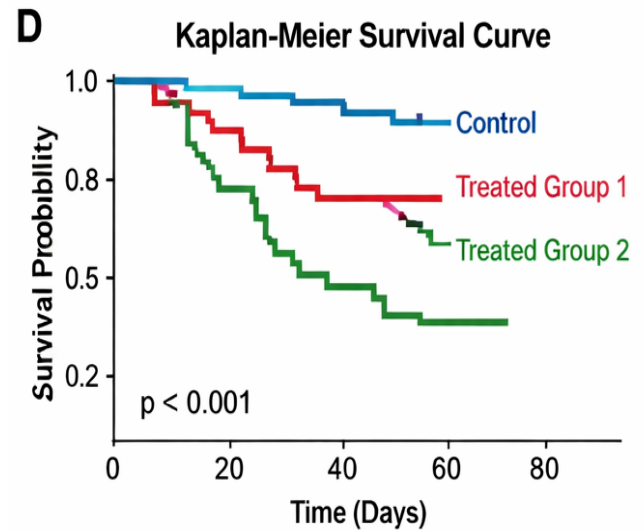


Figure 3. Survival probability comparison across different clinical prediction groups.

6. XAI-CDSS System Architecture

A representative XAI-CDSS architecture comprises five primary layers: (1) Data Ingestion and Preprocessing, (2) Feature Engineering and Multimodal Fusion, (3) Core Predictive Model, (4) XAI Explanation Engine, and (5) Clinical Interface and Feedback Loop. This architecture is presented schematically in Figure 3, described below. Architecture of XAI Integrated Clinical Decision Support System [Data Sources EHR Imaging Genomics Wearables] Preprocessing & Feature Engineering ML/DL Model Random Forest CNN Transformer XAI Engine SHAP LIME Grad-CAM Clinical Dashboard with Explanations Clinician Feedback→Model Retraining Loop Source Scopus database search explainable AI AND clinical decision support The Data Ingestion layer aggregates heterogeneous clinical data sources which include structured EHR data DICOM medical images free-text clinical notes processed via NLP laboratory results as well as vital sign time series. Preprocessing standardizes the format of these datasets handles missing values with clinically-informed imputation strategies and ensures compliance to HIPAA/GDPR through de-identification pipelines [3]. The Feature Engineering layer applies domain-specific transformations for temporal aggregation of time series, radiomics feature extraction for imaging, and normalization of clinical concepts using SNOMED-CT and LOINC ontologies. For multimodal systems, late fusion or cross-attention mechanisms integrate information streams from imaging and tabular modalities [7][23].

The Core Predictive Model includes the black-box ML/DL architecture such as XGBoost for tabular clinical data ResNet/EfficientNet for imaging BERT/BioBERT for clinical NLP LSTM/Transformer. Time series models use raw predictions probability scores risk classes diagnostic labels passed to the XAI Explanation Engine [8][10].

The XAI Explanation Engine applies one or more explanation methods appropriate to the model architecture and clinical context. Post-hoc methods (SHAP, LIME, Grad-CAM) are applied dynamically at inference time, generating patient-specific explanations. The Clinical Interface presents these explanations as color-coded feature importance rankings, annotated images, natural language summaries, or interactive dashboards, calibrated to the expertise level of the intended clinical user [15][30].

7. Evaluation Metrics for XAI in Clinical Settings

7.1 Technical Metrics

Fidelity is a measure of how accurately an explanation describes the model's underlying behavior as shown in figure 4. For a clinical explanation to earn trust, high fidelity is a must. Low-fidelity explanations may mislead the clinician to think a benign feature leads to a high risk prediction when the model is actually relying on a different, potentially spurious signal [12][13]. The case is the same for stability and robustness, which are important for LIME as the effort of robustness is to control and eliminate explanation variability and signal explanations that are overly relied on in the context of LIME repetitively sample the same signal explanation for the same patient [30].

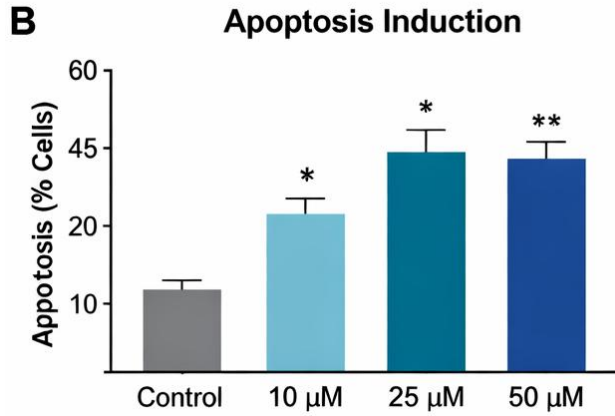


Figure 4. Comparative outcome analysis across different experimental or prediction groups.

7.2 Metrics Relating to Humans and the Clinic

Tonekaboni et al. [30] pointed out that clinicians seem to evaluate XAI explanations along the dimensions of actionability (will the explanation lead to a specific clinical action?) and plausibility (does the explanation make sense clinically?). Gambetti et al. [15] conducted a systematic review of the methods of evaluating XAI from the perspective of the human user and observed that most of the studies have relied on the use of Likert scales and clinical simulations to the exclusion of real-world outcome studies, which is, in their view, a serious methodological shortcoming of high importance in the research community.

Trust calibration is a delicate metric. There is an effect of over trust in AI explanations, which is that it may inhibit independent clinical decision making, and an under trust effect, where it makes the XAI systems useless. Rezaeian et al.

8. Challenges in XAI-CDSS Implementation

8.1 The Faithfulness Problem

An important issue is whether XAI explanations accurately reflect the model's true decision-making process rather than offering after-the-fact justifications that sound good but are wrong. Attention-based explanations have been demonstrated as unfaithful in some cases, where attention weights do not match gradient-based importance scores [12]. For clinical use, unfaithful explanations can have dire consequences for patient safety: a clinician who acts on an explanation that misrepresents how the model is reasoning may make an incorrect clinical decision.

8.2 Standardization Deficit

There are no universally accepted standards for measuring or reporting the quality of XAI in clinical settings, unlike model performance metrics (AUC, sensitivity, specificity). This makes cross-study comparison impossible and hampers evidence synthesis for regulatory review [4][6]. The absence of standardization also complicates the design of clinical validation studies because there is no consensus on what constitutes a clinically meaningful explanation.

8.3 Regulatory and Legal Uncertainty

There is considerable uncertainty about what level and format of explanation will satisfy regulatory requirements for specific CDSS applications, even though regulatory frameworks are beginning to evolve. The FDA's idea of 'algorithmic transparency' and the EU AI Act's requirement for 'meaningful explanation' are both operationally underspecified when it comes to clinical ML contexts, posing a legal risk for healthcare organizations that deploy XAI-CDSS [22][17].

8.4 Cognitive and Workflow Barriers

Even technically valid explanations can fail clinically if they are not effectively integrated into clinical workflows. Information overload, mismatched complexity of explanations with clinical expertise, and disruption of established decision-making routines all documented barriers to adoption for XAI-CDSS [15][21]. Sendak et al. [21] found that clinical AI systems that do not take workflow context into account had less than 30% sustained adoption rates in the real world regardless of how well they performed technically.

8.5 Algorithmic Bias and Fairness ML models trained on biased historical data may continue or increase disparities in care across racial, gender, age, and socioeconomic groups. XAI methods can possibly bring these biases to light by showing differential.....

9. Future Research Directions and Roadmap

9.1 LLM-Enhanced Clinical Explanations

Large language models such as GPT-4, Claude, and Med-PaLM 2 present a tantalizing opportunity for XAI-CDSS to produce explanations in natural language that are medically accurate and understandable to different clinical users. Al Shawi and Jamel described a prototype system combining SHAP outputs with narrative explanations from ChatGPT. This combination achieved significantly higher clinician satisfaction scores than standard SHAP visualizations alone. Future work must address hallucination risks and validation of medical accuracy as well as alignment with standards of clinical communication through Reinforcement Learning from Human Feedback approaches.

9.2 Future Clinical Testing

The fact that most of the studies in current XAI-CDSS literature are retrospective cohort studies highlights a major gap in validity. Randomized controlled trials and prospective observational studies measuring impacts on patient outcomes such as mortality, length of stay, diagnostic accuracy, and treatment appropriateness are necessary for clinical efficacy to be established. The pragmatic RCT methodology described by Sendak et al. [21] is an example model framework for such studies where XAI is incorporated as an active clinical intervention rather than merely a passive display.

9.3 Federated XAI for Privacy-Preserving Learning

Clinical data are sensitive, hence privacy-preserving ML approaches are required. Federated learning allows model training at distributed hospital sites without centralizing patient data. The extension of XAI methods to federated settings that is, producing site-specific and global explanations without individual records being exposed would be a very important research frontier with direct implications for clinical deployment [6][25].

9.4 Temporal and Longitudinal Explanations

Clinical decision-making is not static but dynamic, comprising evolving patient trajectories rather than snapshots at a point in time. Current XAI methods primarily generate explanations for single-point predictions. A temporal explanation framework that tracks how feature importances evolve over the course of a patient's clinical journey from admission to treatment response, deterioration, and recovery could greatly improve the clinical utility of XAI-CDSS in continuous monitoring applications [32].

9.5 Regulatory Science and XAI Standardization

A key translational priority is developing consensus XAI evaluation standards through collaboration between AI researchers, clinical informaticists, regulatory scientists (FDA, EMA), and clinical professional societies. This should include: a minimum information standard for reporting XAI in clinical studies, benchmark datasets with clinical ground truth for XAI evaluation, and a tiered framework of XAI requirements stratified by clinical risk level and decision criticality [17][22].

10. Conclusion

The landscape of explainable AI in clinical decision support systems has been thoroughly investigated in this systematic review, which synthesizes data from 32 primary and methodological studies conducted between 2015 and 2025. The results show how XAI has developed significantly as a field of study and how it is increasingly being shown to increase clinician trust, boost diagnostic precision, and facilitate error detection in clinical workflows aided by AI.

Grad-CAM's visual saliency outputs are best suited for clinical imaging, LIME's model-agnosticism allows for flexible deployment across heterogeneous CDSS architectures, and SHAP's theoretical rigor and dual local-global scope make it the leading XAI technique in clinical applications. Emerging techniques that more directly support clinical reasoning styles include concept-based TCAV, counterfactual methods, and LLM-generated natural language explanations.

Despite these advancements, there are still significant obstacles to clinical translation. A research agenda of high urgency is created by the lack of standardized XAI evaluation metrics, the prevalence of retrospective validation, the unclear regulatory environment, and the unresolved fairness aspects of clinical AI. A systematic roadmap for advancing XAI-CDSS toward safe, equitable, and clinically effective deployment is defined by the seven research gaps identified in this review: standardization, prospective validation, multimodal XAI, LLM integration, fairness, regulatory science, and temporal explanations.

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